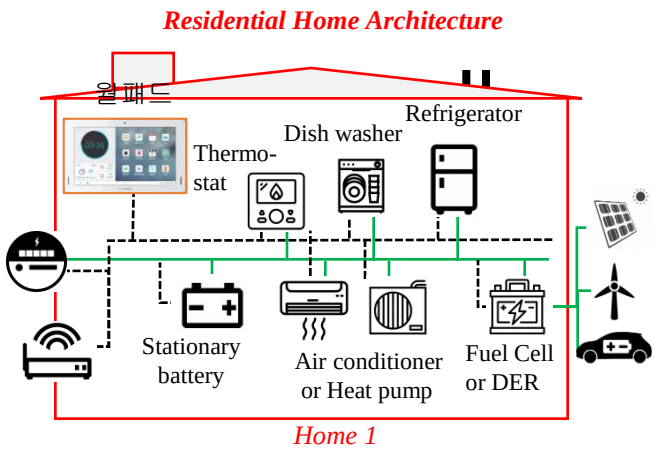
# What is Computer Network?
# What is Internet?

- computer network = a network that allows computers to exchange data (Service & Applications).
- Network=a collection of nodes and links that are connected so as to enable communication between the hosts.
- **Internet?** a specific computer network that provides services to distributed applications, [Network of network, store-and-forward, Statistical Multiplexing, Packet-switched datagram network, stateless architecture, TCP/IP protocol]
  - Service: the function provided by a server (computing) [wiki]
  - Application: software designed to help the user **to perform specific tasks**
- Internet Architecture ~ *Infrastructure* Or Service Interfaces (APIs)?
  - Nut& bolt [technical view]
  - provides services to Distributed applications [service *view*]

# Internet Architecture?

## Please check Dictionary and memorize these terminologies!!

- Architecture=The fundamental organization of a system, embodied in its components, their relationships to each other and the environment, and the principles governing its design and evolution. according to ISO/IEC 42010:20071:

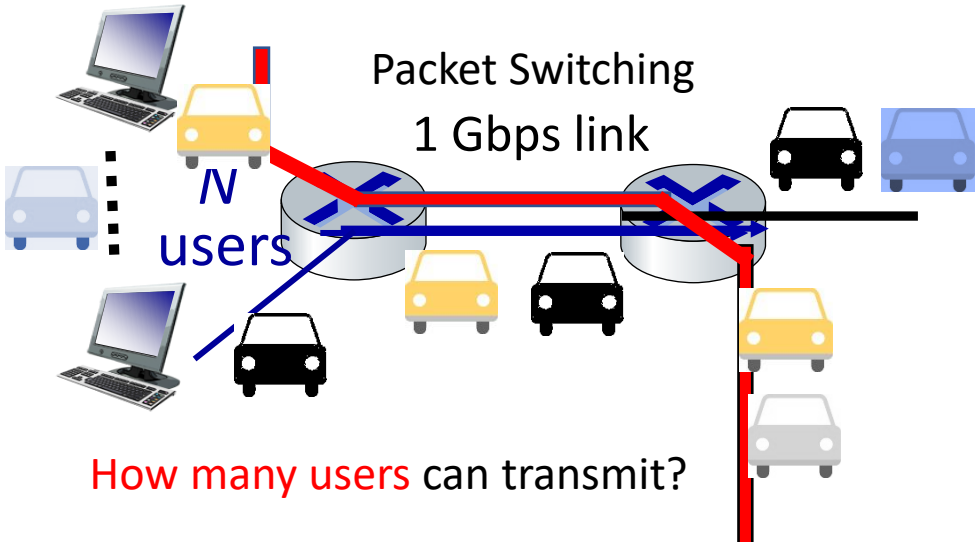


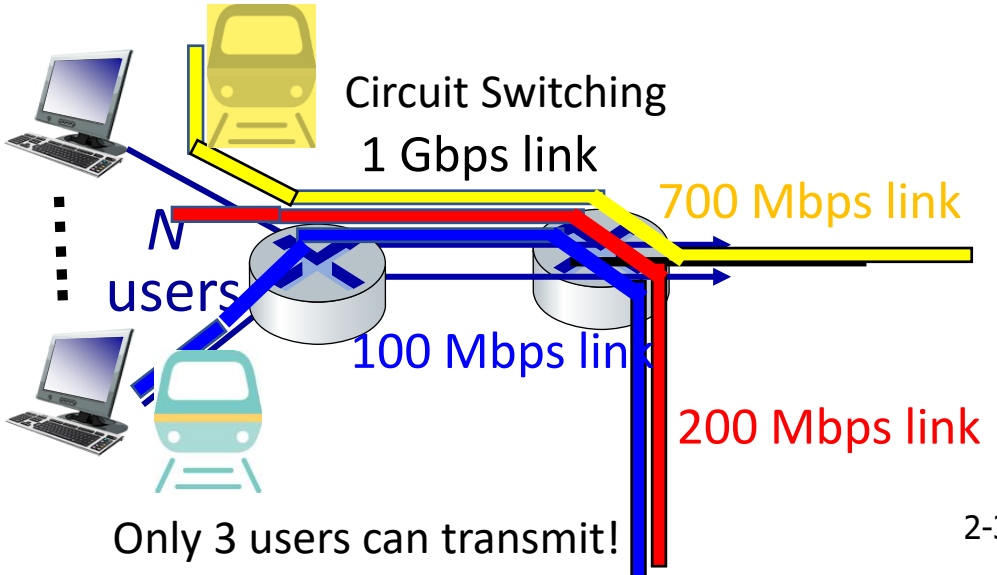

Internet Architecture?

# 1.3.1 Packet-switching (pp.51)

## Store-and-Forward Transmission cf. Circuit switching

- 1.3.1 Packet Switching
- **Packet Switching= a method of exchanging packets that may be shared by multiple simultaneous communication sessions based on packet-by-packet**
- Store-and-Forward Transmission: Most packet switches use store-and-forward transmission at the inputs to the links. Store-and-forward transmission means that the packet switch must receive the entire packet before it can begin to transmit the first bit of the packet onto the outbound link..



# Terminology 1

- computer network = a network that allows computers to exchange data.
- Network=a collection of nodes and links that are connected so as to enable communication between the hosts. {nodes+links} {Hardware+software+cables}
  - Routers and packet switches are both nodes. link refers to the physical path that transfers data from one device to another. Topology refers to the physical or logical arrangement of a network or connectivity
  - Data communications are the (transfer of data) from one device to another via some form of (transmission medium)
- **Internet**=a specific computer network that provides services to distributed applications, [TCP/IP, Network of network, Packet-switched datagram network, Store and forward architecture]
  - End system=host that run application programs at the network edge
  - access network=the network that physically connects an end system to the first router
  - Core network= backbone network provides paths for the exchange of information between different access-networks
  - Media (or medium)= any material substance which can propagate waves or energy
- **Internet architecture**: The fundamental architectural feature of the Internet is the use of **Datagram, the specialized packet to be forwarded to the destination, which is transported across the underlying networks**. In this case, the packet switch provides Store and forward packet switching, that the packet switch must receive the entire packet before it can begin to transmit the first bit of the packet onto the outbound link. The final aspect of this fundamental goal was the assumption of the particular technique for Internetworking these networks as "network of networks" of interconnected ISPs,.

# What is information?

- **Information** or **info** is the resolution of uncertainty, or a collection of related data or knowledge about a topic. **Information** may also refer to:
  - Data, or data used in computing
  - Physical information either of, or contained in a physical system
  - Information technology, the means of processing, storage, or transmission of data
  - Information theory, the mathematical theory of Information and communication
  - Directory assistance, a phone service used to find out a specific telephone number and/or address of a residence, business, or government entity (see Discrete Math)
- Data
  - **Data** are characteristics or information, usually numerical, that are collected through observation.[1] In a more technical sense, data are a set of values of qualitative or quantitative variables about one or more persons or objects, while a **datum** (singular of data) is a single value of a single variable.[2]
- Packet is a chunk of data that can be easily handle. {It consists of header +data+tail}.
  - Frame is the stream of bits received from the network layer into manageable units at link layer.
  - Encapsulation means that the whole packet coming from level N is treated as one integral unit for level N-1.
  - A Packet moves from one node (host or router) to the next node in the route via Packet switching with Store-and-forwarding mechanism.

https://en.wikipedia.org/wiki/Information_(disambiguation)

# What is Packet ? (vs. datagram)

- Packet= a chunk of data that can be easily handle. {It consists of header +data+tail}. Frame is the stream of bits received from the network layer into manageable units at link layer. Encapsulation means that the whole packet coming from level N is treated as one integral unit for level N-1.
- A Packet moves from one node (host or router) to the next node in the route via Packet switching with Store-and-forwarding mechanism.
- Datagram: the specialized packet to be autonomously forwarded to the destination, which is transported across the underlying networks.
- Packet Switching= a method of exchanging packets that may be shared by multiple simultaneous communication sessions based on packet-by-packet **multiplexing.**
- Circuit switching=a method of exchanging data in which two network nodes establish a dedicated communications channel (circuit) through the network.
- **Statistical multiplexing=** "on-demand" link sharing service based on packet rather than multiplexing that pre-allocates resources for each data stream
  - Multiplexing = the process of combining signals from multiple sources for transmission across a single shared data link
- In this case, the packet switch follows Store and forward packet switching, i.e., the relaying of packets from incoming port to outgoing port. Store and forward means that the packet switch must receive the entire packet before it can begin to transmit the first bit of the packet onto the outbound link.

# Summary: delay, loss, throughput in networks

- **Network delay** is a design and performance characteristic of a telecommunications network. It specifies the latency for a bit of data to travel across the network from one communication endpoint to another. It is typically measured in multiples or fractions of a second. Delay may differ slightly, depending on the location of the specific pair of communicating endpoints. Engineers usually report both the maximum and average delay, and they divide the delay into several parts:
  - Processing delay – time it takes a router to process the packet header
  - Queuing delay – time the packet spends in routing queues
  - Transmission delay – time it takes to push the packet's bits onto the link
  - Propagation delay – time for a signal to reach its destination
- In general terms, **throughput** is the rate of production or the rate at which something is processed.
  - Throughput=the number of bits that can pass through a point in one second. Throughput is usually measured in bits per second (bit/s or bps), and sometimes in data packets per second (p/s or pps) or data packets per time slot.
  - Transmission rate= the number of bits sent per second
  - Bandwidth= the rate of data transfer that can pass through a point in one second
- Others
  - Packet loss - packet may be dropped with limited buffer or corrupted by error (aka lost)
  - Jitter= a phenomenon in real-time traffic caused by gaps between consecutive packets (or data) at the receiver

https://en.wikipedia.org/wiki/

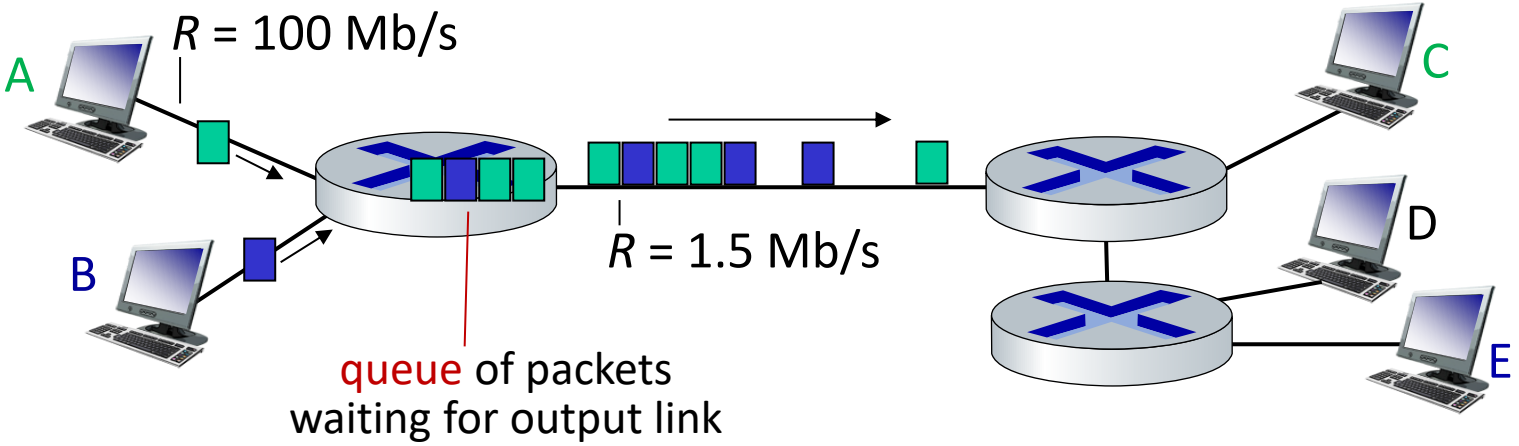
# Summary: Strength, Weak, opportunity, and threat (SWOT) (Packet switching versus circuit switching)

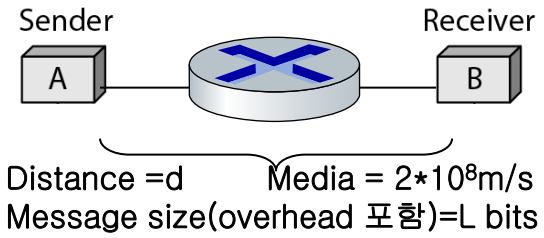
| SWOT | Strength | Weak | opportunity | threat |
|---|---|---|---|---|
| Packet switching | Availability, efficiency (statistical multiplexing gain) | Delay uncertainty | simple and easy access [deployment] (burst traffic) | Excessive Congestion |
| circuit switching | Assurance (periodic traffic) | Connection setup (reservation), | Guarantee QoS [delay, throughput] | Low utilization (no sharing) |

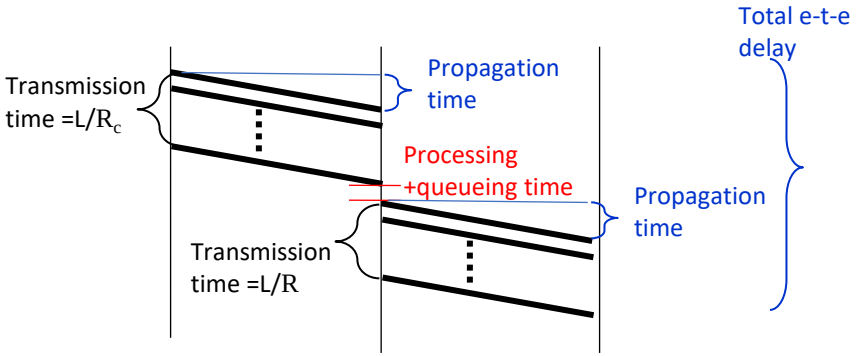
**Packet Switching:** a method of exchanging packets that may be shared by multiple simultaneous communication sessions based on packet-by-packet
**Store-and-forward transmission: that** the packet switch must receive the entire packet before it can begin to transmit.
**Statistical multiplexing:** "on-demand" link sharing service based on packet rather than multiplexing that pre-allocates resources for each data stream
Congestion: An excess of traffic when a network node or link is carrying more data than it can handle

Availability: The probability that an item will operate satisfactorily at a given point, the ability of a system to provide the desired service.
Assurance: The act of assuring; a declaration tending to inspire full confidence; something designed to give confidence.
Utilization: the rate of usage, The state of being used.

# Summary: Packet-switching: queueing delay, loss




*Packet queuing and loss:* if arrival rate (in bps) to link exceeds transmission rate (bps) of link for a period of time:

- packets will queue, waiting to be transmitted on output link
- packets can be dropped (lost) if memory (buffer) in router fills up (<u>no place to store</u> <u>Because the queue capacity is finite</u>).

# Summary: delay, loss, throughput in networks

- **Network delay** is a design and performance characteristic of a telecommunications network. It specifies the latency for a bit of data to travel across the network from one communication endpoint to another. It is typically measured in multiples or fractions of a second. Delay may differ slightly, depending on the location of the specific pair of communicating endpoints. Engineers usually report both the maximum and average delay, and they divide the delay into several parts:
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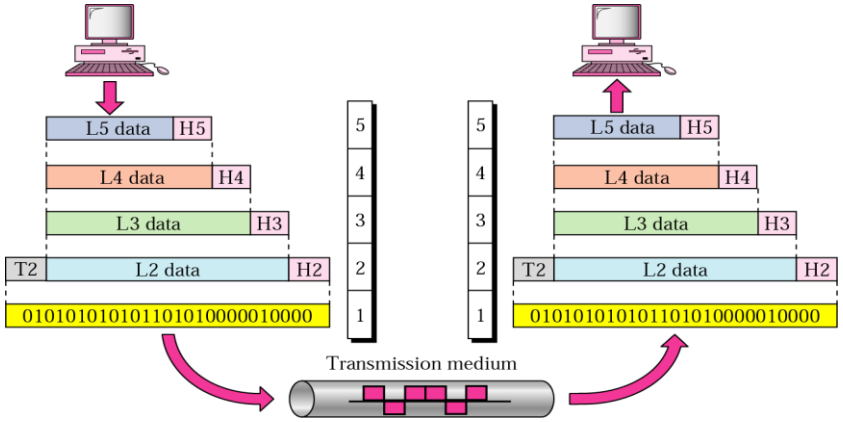
https://en.wikipedia.org/wiki/

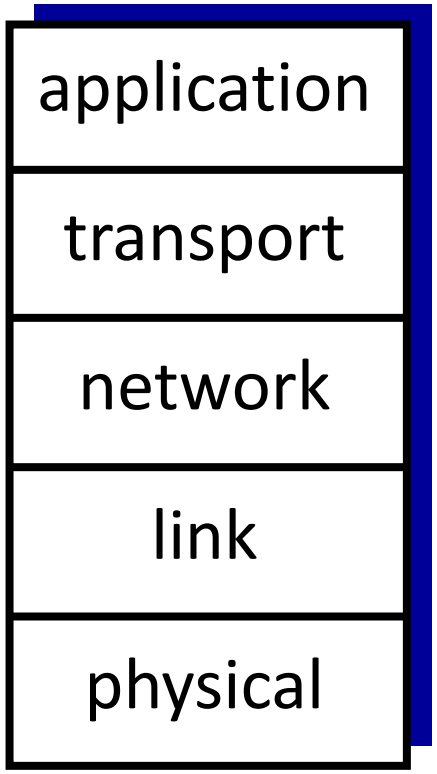
# Summary: Layering

- Key technique to implement communication protocols; provides
  - Modularity
  - Abstraction
  - Reuse
- Key design decision:
  what functionality to put in each layer?
- The result of Layering:
  Encapsulation



# Summary of Layering: Internet protocol stack

- *application:* supporting network applications
  - IMAP, SMTP, HTTP
- *transport:* process-process data transfer
  - TCP, UDP
- *network:* routing of datagrams from source to destination
  - IP, routing protocols
- *link:* data transfer between neighboring network elements
  - Ethernet, 802.11 (WiFi), PPP
- *physical:* bits "on the wire"



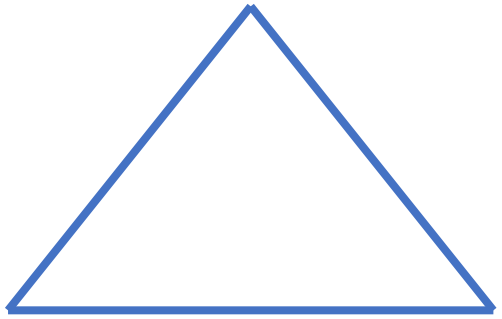
# Summary: End-to-End Arguments

- If the application can do it, don't do it at a lower layer -- anyway the application knows the best what it needs -- "as Arguments"
  - add functionality in lower layers iff it is (1) used and improves performances of a large number of applications, and (2) does not hurt other applications
- Key design decision: peer-to-peer protocol
  what is the role of End-to-End argument in each layer?
- Success story: Internet

# Summary of Design Goals of The Internet Protocol

- Challenge of building a good (network) system: find the right balance between:

Reuse, implementation effort (apply layering concepts)
Easy to reuse vs. optimized implementation



Performance (packet switching)
Best effort vs. QoS guarantee

end-to-end argument
Centralized vs. distributed

- No universal answer: the answer depends on the goals and assumptions!

- Is there any other design goals?
- What do you think? →

# Terminology 3

- Protocol= A set of (rules) that governs (communications/data delivery ).
- **Protocol** defines the (format) and the (Order) of messages exchanged between two or more communicating entities, as well as the (Actions) taken on the transmission and/or receipt of a message or other event.
  - (Peer-to-peer) protocol = A process on a sending and a receiving machine that communicates at a (given layer)
  - The network do not maintain any information about the clients as stateless protocol.
- Protocol Layering
  - Layering is an easy technology dealing with complex systems
  - Encapsulation means that the whole packet coming from level N is treated as one integral unit for level N-1.
- Internet Protocol Layering
  - What are the five layers in the Internet protocol stack? Application, Transport, Network, Link, Physical.
  - Application Layer is where network applications and their application-layer protocols reside. The Internet's Application Layer includes many protocols such as HTTP Protocol (which provides for Web document request and transfer)), SMTP which provides for the transfer of e-mail messages, and FTP which provides for the transfer of files between two end systems.
- Why we need Protocols? For interacting and exchanging messages among different systems (e.g., human or ?)!
- **Why does the protocol need standard**? Make consensus (e.g., Internet, 5G, 6G) among different vendors cf. de-factor (e.g., Apple, google) for interoperability

# Summary Chapter 1 Computer Networks and the Internet

- **computer network ? a network that allows computers to exchange data.**
- Network? a collection of nodes and links that are connected so as to enable communication between the hosts [=End systems]. {nodes+links} {Hardware+software+cables}
- **Internet?** a specific computer network that provides services to distributed applications, [TCP/IP, Network of network, Packet-switched datagram network, stateless architecture ]
- Architecture=The fundamental organization of a system, embodied in its components, their relationships to each other and the environment, and the principles governing its design and evolution. according to ISO/IEC 42010:20071:
- **Infrastructure** is the set of facilities and systems that serve a country, city, or other area,[1] and encompasses the services and facilities necessary for its economy, households and firms to function.[2]
- Frameworks is a collection of classes and tools that help you developing great software.
- Packet? a chunk of data that can be easily handle {header+data+tail}
- **Packet Switching? a method of exchanging packets that may be shared by multiple simultaneous communication sessions based on packet-by-packet**
- Protocol? a set of rules that **that governs communications/data delivery.**
  - *defines the format and the order of messages exchanged between two or more communicating entities, as well as the actions taken on the transmission and/or receipt of a message or other event.*

# Terminology 1 in chapter1

- **computer network = a network that allows computers to exchange data.**
- **Network=a collection of nodes and links that are connected so as to enable communication between the hosts.** {nodes+links} {Hardware+software+cables}
- **Internet=a specific computer network that provides services to distributed applications,**
  [TCP/IP, Network of network, Packet-switched datagram network, stateless architecture ]
- Packet= a chunk of data that can be easily handle **{header+data+tail}, Datagram : the specialized packet to be autonomously forwarded to the destination, which is transported across the underlying networks.**
- **Protocol = a set of rules that that governs communications/data delivery.**
  - ***defines the format and the order of messages exchanged between two or more communicating entities, as well as the actions taken on the transmission and/or receipt of a message or other event.***
- End system = host that run application programs at the network edge
- access network = the network that physically connects an end system to the first router
- Core network= backbone network provides paths for the exchange of information between different access-networks
- Media (or medium)= any material substance which can propagate waves or energy
- Link refers to the physical path that transfers data from one device to another
- Transmission rate= the number of bits sent per second
- Bandwidth**=** the rate of data transfer **that can pass through a point in one second**
- Multiplexing = the process of combining signals from multiple sources for transmission across a single shared data link

# Terminology 2 in chapter1

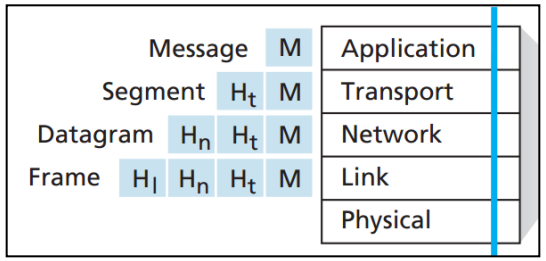
- **Store-and-forward= means that the packet switch must receive the entire packet before it can begin to transmit the first bit of the packet onto the outbound link.**
- **The network do not maintain any information about the clients as stateless protocol.**
- **Packet Forwarding = the relaying of packets from incoming port to outgoing port via Packet switching**
- **Routing**=the process of selecting best paths in a network
- **(Router) was a fundamental component of the packet switching in Internet architecture.**
- Delay (time)=transmission delay, propagation delay, queuing delay
  - Propagation delay= **time required for a signal to travel from one point to anther**
  - **transmission delay** (or **store-and-forward delay)=** the amount of time required to push all the packet's bits into the wire
  - **queuing delay** = the time a job waits in a queue until it can be executed
  - Processing delay= the time it takes routers to process the packet header
- **Packet Switching= a method of exchanging packets that may be shared by multiple simultaneous communication sessions based on packet-by-packet**
- Circuit switching=a method of exchanging data in which two network nodes establish a dedicated communications channel (circuit) through the network
- **(Topology) = refers to the physical or logical arrangement of a network or connectivity**
- **Statistical multiplexing= "on-demand" link sharing service based on packet rather than multiplexing that pre-allocates resources for each data stream**

# Terminology 3 in chapter1

- **(interconnecting) the networks as "network of networks" of interconnected Internet Service Providers (ISPs).**
- Layering: A technique to organize a network system into a succession of logically distinct entities, such that the service provided by one entity is solely based on the service provided by the previous (lower level) entity
- Encapsulation= For level N-1, the whole packet coming from level N is treated as one integral unit.
- End-to-End Arguments: A technique for two processes communicating with each other via some communication means
- (Peer-to-peer) protocol = A process on a sending and a receiving machine that communicates at a (given layer)
- Data communications are the (transfer of data) from one device to another via some form of (transmission medium)
- (Collision)= The event that occurs when two transmitters send at the same time on a (channel) designed for only one transmission at a time; data will be destroyed.
- (Frame): the stream of bits received from the network layer into manageable units at link layer
- (Propagation delay [time])= time required for a signal to travel from one point to anther
- Transmission rate= the number of bits sent per second
- (Transparency) = the ability to send (any bit pattern) as data without it being mistaken for control bits
- Throughput=the number of bits that can pass through a point in one second
- Jitter= a phenomenon in real-time traffic caused by gaps between consecutive packets (or data) at the receiver
- Piggybacking = the inclusion of acknowledgement on a data frame

# Terminology 4 in chapter1

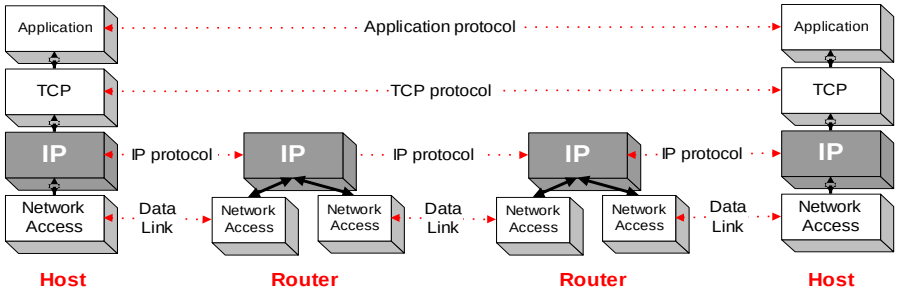
- Internet Service Providers (ISPs): ISPs provide a variety of types of network access to the end systems.
- Internet Application Programming Interface (API): It is a set of rules that the sending program must follow so that the Internet can deliver the data to the destination program.
- Access Networks: An access network is a user network that connects subscribers to a particular service provider and, through the carrier network, to other networks such as the Internet [1].
- Packet Loss: either the arriving packet or one of the already-queued packets will be dropped.
- Forwarding Table (Router): Each router has a forwarding table that maps destination addresses (or portions of the destination addresses) to that router's outbound links.
- Internet exchange points (IXPs): It is a meeting point where multiple ISPs can peer together.
- Delays = nodal processing delay + queuing delay + transmission delay + propagation delay
- Instantaneous throughput (at any instant of time) vs. average throughput



# Review Questions: Design Goals of Internet Protocol

- What is Store & forward packet switching?
- Datagram?
- What is Layered Architecture in designing the Internet protocol?
- What is Encapsulation?
- What is the differences between Point-to-point network and broadcast network?
- What is Network Topology? And explain with an example.
- Why we need Protocols?
- **Why does the protocol need standard**?

# Review Questions: Design Goals of Internet Protocol




- What are Design Goals of Internet Protocol?
- What is Layering?
- Why do you need Layering?
- What is End-to-End Arguments? of peer-to-peer protocol
- Why do you need End-to-End Arguments?

# Ch1 Review Questions

- R13. Why is it said that packet switching employs statistical multiplexing?
- R22. List five tasks that a layer can perform.
- R23. What are the five layers in the Internet protocol stack? What are the principal responsibilities of each of these layers?
- R24. Which layers in the Internet protocol does a router process?
- R25. What is an application-layer message? A transport-layer segment? Network-layer datagram? A Link-layer frame?
- R26.What is the difference between a virus, a worm, and a Trojan horse?

# Summary in Chapter 2

- Internet provides 5-layered protocol architecture, which consists of ( , , , , ) . The layered protocol process of passing a packet from a higher layer to a lower layer enables (C. Encapsulation), in which each layer simply appends additional information called (D. header) that needs to process the packet.
  - Application, Transport, Network, Link, Physical.
- Internet is the infrastructure that provides services to distributed (A. Application). This infrastructure provides programming interface to applications. An application process running on one host identifies a process running on another host by (B. socket) API interface with IP address and port number of the destination process.
- Application Layer is where network applications and their application-layer protocols reside. The Internet's Application La yer includes many protocols such as HTTP Protocol (which provides for Web document request and transfer)), SMTP whic h provides for the transfer of e-mail messages, and FTP which provides for the transfer of files between two end systems.
  - Particularly, the (HTTP) protocol only provides the functions to request and retrieve files (e.g., Web pages or (H. HTML, or Web pages files). The protocol is originally (I. stateless) since it does <u>not keep any information about the clients</u> during communication sessions. However, it maintains "state" by using (J. cookies) that allow websites to track users across different sessions when the browser is closed and re-opened.
  - Web (K. caching or proxy) can bring the desired content "closer" to the user to reduce the delay, possibly to the same LAN to which the user's host is connected. It is a network entity that satisfies HTTP requests on the behalf of an origin Web server with its own disk storage and copies of recently requested objects in the storage.

# Review: Chapter 2

- Network application=Application programs that run on different end systems and communicate with each other over the network.
  - Application will be a great service to humanity, or will please your professor, or will bring you great wealth, or will simply be fun to develop.
  - In a client-server architecture, there is an always-on host, called the server, which services requests from many other hosts, called clients. the server has a fixed, well-known address, called an IP address and port number.
  - In a P2P architecture, the application exploits direct communication between pairs of intermittently connected hosts, called peers.
- The **HyperText Transfer Protocol (HTTP) is** the Web's application-layer protocol defined in [RFC 1945] and [RFC 2616].
  - (HTTP) protocol only provides the functions to request and retrieve files (e.g., Web pages or (H. HTML, or Web pages files).
  - Because an HTTP server maintains no information about the clients, HTTP is said to be a **stateless protocol**.
- A **Web page** (also called a document) consists of objects. An **object** is simply a file—such as an HTML file, a JPEG image, a Java applet, or a video clip—that is addressable by a single URL.
  - http://www.someSchool.edu/someDepartment/picture.gif
- Web Caching (called a proxy server)=a network entity that satisfies HTTP requests on the behalf of an origin Web server.
- the **round-trip time (RTT)** is the time it takes for a small packet to travel from client to server and then back to the client. The RTT includes packet-propagation delays, packet queuing delays in intermediate routers and switches, and packet-processing delays.
- multi-object, pipelined HTTP requests over single TCP connection
- HTTP over UDP Why not?

[wiki] cache is a hardware or software component that stores data so that future requests for that data can be served faster; the data stored in a cache might be the result of an earlier computation or a copy of data stored elsewhere.

# Terminology 1 in Chapter 2

- **Network application=Application programs that run on different end systems and communicate with each other over the network.**
  - Application will be a great service to humanity, or will please your professor, or will bring you great wealth, or will simply be fun to develop.
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  - **Internet is still a stateless architecture, isn't it?**
- A **Web page** (also called a document) consists of objects. An **object** is simply a file—such as an HTML file, a JPEG image, a Java applet, or a video clip—that is addressable by a single URL.
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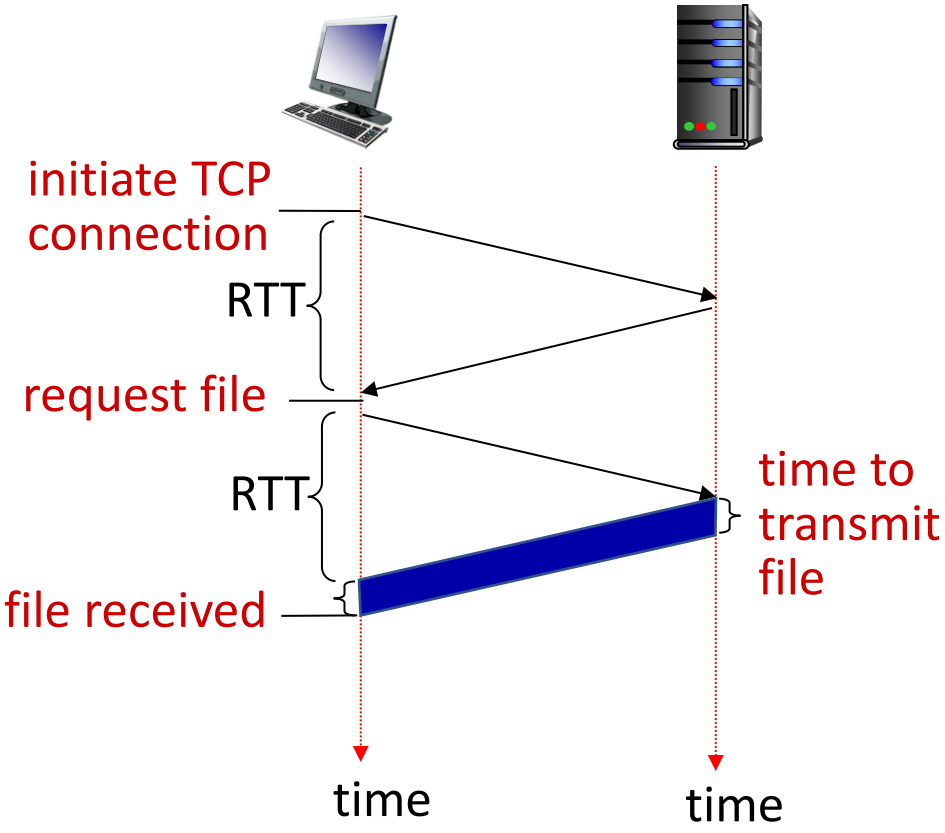
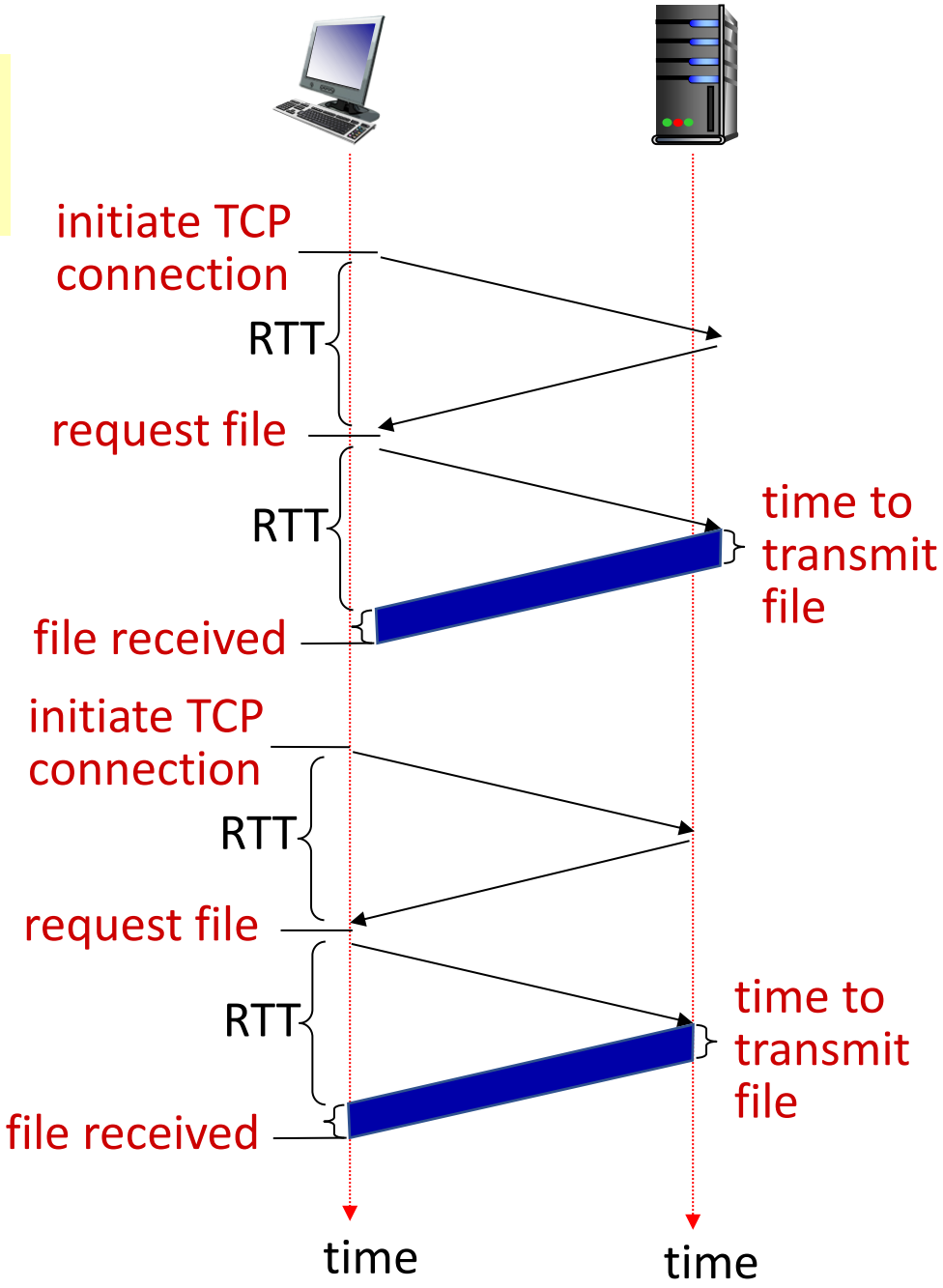
- Response time: **time for a process initiates a request until receives the response from server back!**

# Summary: Non-persistent HTTP



# Chapter 2: Summary

our study of network application layer is now complete!

- application architectures
  - client-server
  - P2P
- application service requirements:
  - reliability, bandwidth, delay
- Internet transport service model
  - connection-oriented, reliable: TCP
  - unreliable, datagrams: UDP
- specific protocols:
  - HTTP
  - SMTP, IMAP
  - DNS
  - P2P: BitTorrent
- video streaming, CDNs
- socket programming: TCP, UDP sockets

# Chapter 2: Summary

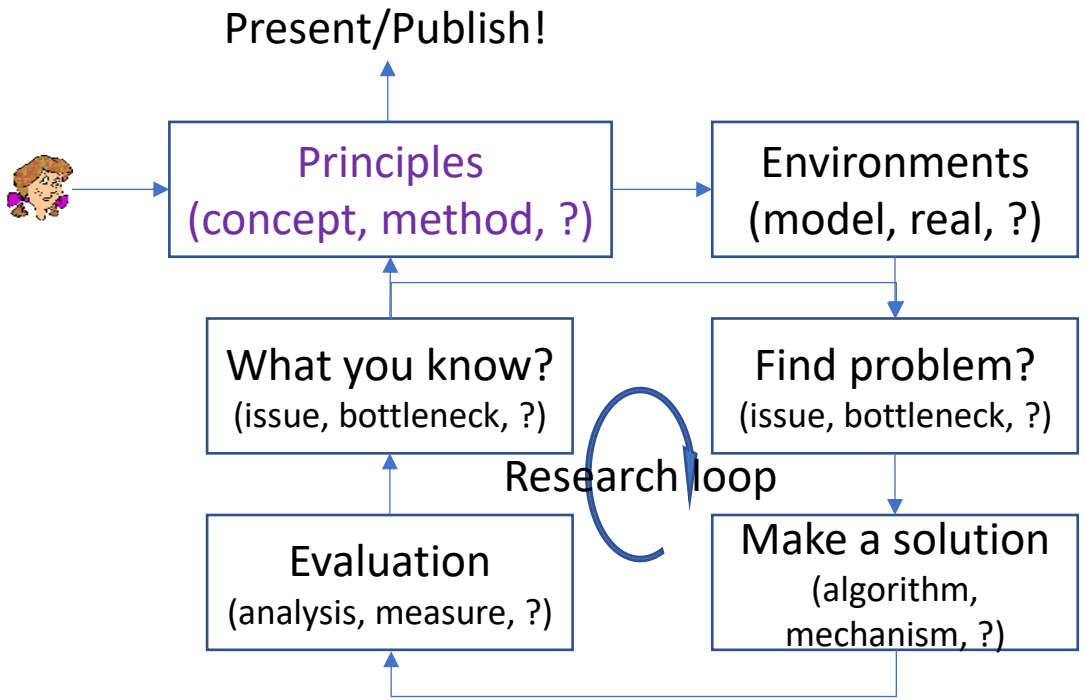
## Most importantly: learned about *protocols*!

- typical request/reply message exchange:
  - client requests info or service
  - server responds with data, status code
- message formats:
  - *headers*: fields giving info about data
  - *data:* info(payload) being communicated

important themes:

- centralized vs. decentralized
- stateless vs. stateful
- scalability
- reliable vs. unreliable message transfer
- "complexity at network edge"

# Ch2 Summary




2.1 principles of network applications

2.2 Web and HTTP

2.3 FTP electronic mail : SMTP, POP3, IMAP

2.4 DNS

2.5 P2P applications

2.6 CDN: video streaming and content distribution networks

2.7 socket programming with UDP and TCP

we've studied the conceptual and the implementation aspects of network applications. We've learned about the ubiquitous client-server architecture adopted by many Internet applications and seen its use in the HTTP, FTP, SMTP, POP3, and DNS protocols. We've studied these important application-level protocols, and their corresponding associated applications (the Web, file transfer, e-mail, and DNS) in some detail.

We've walked through the use of sockets for connection-oriented (TCP) and connectionless (UDP) end-to-end transport services. The first step in our journey down the layered network architecture is now complete!

# 2.8 Summary

- In this chapter, we've studied the conceptual and the implementation aspects of network applications. We've learned about the ubiquitous client-server architecture adopted by many Internet applications and seen its use in the HTTP, SMTP, POP3, and associated DNS protocols. We've studied these important application-level protocols, and their corresponding associated applications (the Web, file transfer, e-mail, and DNS) in some detail. We've learned about the P2P architecture and how it is used in many applications. We've also learned about streaming video, and how modern video distribution systems leverage CDNs. We've examined how the socket API can be used to build network applications. We've walked through the use of sockets for connection-oriented (TCP) and connectionless (UDP) end-to-end transport services. The first step in our journey down the layered network architecture is now complete!
- At the very beginning of this book, in Section 1.1, we gave a rather vague, bare-bones definition of a protocol: "the format and the order of messages exchanged between two or more communicating entities, as well as the actions taken on the transmission and/or receipt of a message or other event." The material in this chapter, and in particular our detailed study of the HTTP, SMTP, POP3, and DNS protocols, has now added considerable substance to this definition. Protocols are a key concept in networking; our study of application protocols has now given us the opportunity to develop a more intuitive feel for what protocols are all about.
- In Section 2.1, we described the service models that TCP and UDP offer to applications that invoke them. We took an even closer look at these service models when we developed simple applications that run over TCP and UDP in Section 2.7. However, we have said little about how TCP and UDP provide these service models. In the next chapter we'll take a careful look at not only the what, but also the how and why of transport protocols.

# Ch2. Minute Paper 1

- **1.** List the five types of networking application services mentioned in this chapter.
  **Answer:**
- **2.** Which application services effectively hide the network from the end user?
- **Answer:**
- **3.** Which application services make use of servers to hold and store data, waiting until another user is available in the network? **Answer:**

.

- **4.** Which application service allows a server to present multiple text, graphics, video, audio images, and sounds to the client? **Answer:**
- **5.** In general terms, compare and contrast an enterprise network with the Internet.
- **Answer:**

- **6.** Does the Internet include enterprise networks, home users who connect to the Internet, both, or neither? Explain your answer.
- **Answer:**.

# Ch2. Minute Paper 2

- **7.** Explain the basic difference between a web page and a website. **Answer:**

.

- **8.** What's a network cloud? **Answer:**

- **9.** Compare and contrast enterprise networks and Internet service provider networks in a few sentences. **Answer:**

  .

- **10.** What three-word term refers to the text that you need to type into a web browser to reach a particular website? What popular jargon might be used instead of that term?
- **Answer:**
  .
- **11.** Imagine that Fred connects to the Internet using ISP1, and Barney uses ISP2. When Fred sends Barney an e-mail, to what computer does Fred send the e-mail? If the e-mail is not sent directly to Barney's computer, explain why. (see figure shown in the previous page)
- **Answer:**

# Minute Paper 1 in Chapter 2

- Internet is the infrastructure that provides services to distributed (A. ). This infrastructure provides programming interface to applications. An application process running on one host identifies a process running on another host by (B. ) API interface with IP address and port number of the destination process.
- Application Layer is where network applications and their application-layer protocols reside. The Internet's Application Layer includes many protocol s such as HTTP Protocol (which provides for Web document request and transfer)), SMTP which provides for the transfer of e-mail messages, and FT P which provides for the transfer of files between two end systems.
  - Particularly, the ( ) protocol only provides the functions to request and retrieve files (e.g., Web pages or (H **files**). Web page consists of base ( ) files which includes several referenced objects, each addressable by a URL, e.g., www.someschool.edu/someDept/pic.gif. The Internet's application-layer protocol is originally (I. ) since it does <u>not keep any information about the clients</u> during communication sessions. However, it maintains "state" by using (J. ) that allow websites to track users across different sessions when the browser is closed and re-opened.
  - Web (K. ) can bring the desired content "closer" to the user to reduce the delay, possibly to the same LAN to which the user's host is connected. It is a network entity that satisfies HTTP requests on the behalf of an origin Web server with its own disk storage and copies of recently requested objects in the storage.

- **Application will be a great service to humanity, or will please your professor, or will bring you great wealth, or will simply be fun to develop. ( ) applications are Application programs that run on different end systems and communicate with each other over the network. An application process running on one host identifies a process running on another host by ( ) API interface with IP address and port number of the destination process.**
- **Application Layer is where network applications and their application-layer protocols reside. The Internet's Application Layer includes many protocols such as HTTP Protocol (which provides for Web document request and transfer)), SMTP which provides for the transfer of e-mail messages, and FTP which provides for the transfer of files between two end systems.**
- **Particularly, the ( ) protocol only provides the functions to request and retrieve files (e.g., Web pages). Web page consists of base (HTML ) files which includes several referenced objects, each addressable by a URL, e.g., www.someschool.edu/someDept/pic.gif. The Internet's application-layer protocol is originally stateless since it does not keep any information about the clients during communication sessions. However, it maintains "state" by Web clients using ( ) that allow websites to track users across different sessions when the browser is closed and re-opened.**
- **Web ( ) can bring the desired content "closer" to the user to reduce the delay, possibly to the same LAN to which the user's host is connected. It is a network entity that satisfies HTTP requests on the behalf of an origin Web server with its own disk storage and copies of recently requested objects in the storage.**